

## SAFETY DATA SHEET

Creation Date 13-September-2010

Revision Date 28-December-2021

Revision Number 5

### 1. Identification

**Product Name** 3-Methyl-1-butanol

**Cat No. :** AC451660000; AC451660010

**CAS-No** 123-51-3  
**Synonyms** Isoamyl alcohol; Isopentyl alcohol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                             |            |
|-----------------------------------------------------------------------------|------------|
| <b>Flammable liquids</b>                                                    | Category 3 |
| <b>Acute Inhalation Toxicity</b>                                            | Category 4 |
| <b>Skin Corrosion/Irritation</b>                                            | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                                    | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b>                     | Category 3 |
| Target Organs - Respiratory system.                                         |            |
| <b>Health Hazards Not Otherwise Classified</b>                              | Category 1 |
| Prolonged or repeated contact may dry skin and cause irritation or cracking |            |

#### Label Elements

**Signal Word**  
Warning

**Hazard Statements**

Flammable liquid and vapor  
Harmful if inhaled  
Causes skin irritation  
Causes serious eye irritation  
May cause respiratory irritation  
Prolonged or repeated contact may dry skin and cause irritation or cracking

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting/equipment  
Use only non-sparking tools  
Take precautionary measures against static discharges  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Call a POISON CENTER/ doctor if you feel unwell  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component            | CAS-No   | Weight % |
|----------------------|----------|----------|
| 1-Butanol, 3-methyl- | 123-51-3 | >95      |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

|                                        |                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Get medical attention.                                                                                                                                                                                                                                                               |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                                                                                                                                                                                 |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                        |

## 5. Fire-fighting measures

|                                         |                                                                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                |
| <b>Flash Point</b>                      | 45 °C / 113 °F                                                                                                                          |
| <b>Method -</b>                         | No information available                                                                                                                |
| <b>Autoignition Temperature</b>         | 365 °C / 689 °F                                                                                                                         |
| <b>Explosion Limits</b>                 |                                                                                                                                         |
| <b>Upper</b>                            | 8.0 vol %                                                                                                                               |
| <b>Lower</b>                            | 1.2 vol %                                                                                                                               |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                |

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 2             | 2                   | 0                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.            |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                                                            |
| <b>Methods for Containment and Clean Up</b> | Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Take precautionary measures against static discharges. Use only non-sparking tools.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area. Incompatible Materials. Strong oxidizing agents. Metals. Alkali metals. Halogens. Acids. Acid anhydrides. Acid chlorides. Isocyanates.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component            | Alberta                                                                                    | British Columbia              | Ontario TWAEV                 | Quebec                                                                                     | ACGIH TLV                     | OSHA PEL                                                                                                                                                                         | NIOSH IDLH                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------|-------------------------------|-------------------------------|--------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1-Butanol, 3-methyl- | TWA: 100 ppm<br>TWA: 361 mg/m <sup>3</sup><br>STEL: 125 ppm<br>STEL: 451 mg/m <sup>3</sup> | TWA: 100 ppm<br>STEL: 125 ppm | TWA: 100 ppm<br>STEL: 125 ppm | TWA: 100 ppm<br>TWA: 361 mg/m <sup>3</sup><br>STEL: 125 ppm<br>STEL: 452 mg/m <sup>3</sup> | TWA: 100 ppm<br>STEL: 125 ppm | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 360 mg/m <sup>3</sup><br>(Vacated) STEL: 125 ppm<br>(Vacated) STEL: 450 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 360 mg/m <sup>3</sup> | IDLH: 500 ppm<br>TWA: 100 ppm<br>TWA: 360 mg/m <sup>3</sup><br>STEL: 125 ppm<br>STEL: 450 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material              | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Viton (R) | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                         |
|----------------------------------------|-----------------------------------------|
| Physical State                         | Liquid                                  |
| Appearance                             | Clear                                   |
| Odor                                   | Characteristic                          |
| Odor Threshold                         | No information available                |
| pH                                     | 6.5 25 g/l aq.sol                       |
| Melting Point/Range                    | -117 °C / -178.6 °F                     |
| Boiling Point/Range                    | 130 - 132 °C / 266 - 269.6 °F 760 mm HG |
| Flash Point                            | 45 °C / 113 °F                          |
| Evaporation Rate                       | No information available                |
| Flammability (solid,gas)               | Not applicable                          |
| Flammability or explosive limits       |                                         |
| Upper                                  | 8.0 vol %                               |
| Lower                                  | 1.2 vol %                               |
| Vapor Pressure                         | 4 hPa @ 20 °C                           |
| Vapor Density                          | 3.04 (Air = 1.0)                        |
| Specific Gravity                       | 0.807-0.811                             |
| Solubility                             | miscible                                |
| Partition coefficient; n-octanol/water | No data available                       |
| Autoignition Temperature               | 365 °C / 689 °F                         |
| Decomposition Temperature              | 335 °C                                  |
| Viscosity                              | 4.3 mPa s at 20 °C                      |
| Molecular Formula                      | C5 H12 O                                |
| Molecular Weight                       | 88.15                                   |

## 10. Stability and reactivity

|                                  |                                                                                                               |
|----------------------------------|---------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                    |
| Stability                        | Stable under normal conditions.                                                                               |
| Conditions to Avoid              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.         |
| Incompatible Materials           | Strong oxidizing agents, Metals, Alkali metals, Halogens, Acids, Acid anhydrides, Acid chlorides, Isocyanates |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                                       |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                      |
| Hazardous Reactions              | None under normal processing.                                                                                 |

## 11. Toxicological information

**Acute Toxicity****Product Information****Component Information**

| Component            | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation             |
|----------------------|---------------------------|------------------------------|-----------------------------|
| 1-Butanol, 3-methyl- | LD50 = 5770 mg/kg ( Rat ) | LD50 = 3250 mg/kg ( Rabbit ) | LC50 > 2000 ppm ( Rat ) 8 h |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Irritating to eyes and respiratory system                                                |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component            | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------|----------|------------|------------|------------|------------|------------|
| 1-Butanol, 3-methyl- | 123-51-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** Tumorigenic effects have been reported in experimental animals.

## 12. Ecological information

### Ecotoxicity

This product contains the following substance(s) which are hazardous for the environment. .

| Component            | Freshwater Algae                                                                                         | Freshwater Fish                       | Microtox              | Water Flea                               |
|----------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------|------------------------------------------|
| 1-Butanol, 3-methyl- | EC50: = 493 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: = 181 mg/L, 96h<br>(Desmodesmus subspicatus) | LC50 96 h 700 mg/L<br>(rainbow trout) | EC50 = 2500 mg/L 17 h | EC50: = 260 mg/L, 48h<br>(Daphnia magna) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component            | log Pow |
|----------------------|---------|
| 1-Butanol, 3-methyl- | 1.28    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                             |                      |
|-----------------------------|----------------------|
| <b>UN-No</b>                | UN1105               |
| <b>Proper Shipping Name</b> | PENTANOLS            |
| <b>Technical Name</b>       | (3-METHYL-1-BUTANOL) |
| <b>Hazard Class</b>         | 3                    |

|                             |           |
|-----------------------------|-----------|
| <b>Packing Group</b>        | III       |
| <b>TDG</b>                  |           |
| <b>UN-No</b>                | UN1105    |
| <b>Proper Shipping Name</b> | PENTANOLS |
| <b>Hazard Class</b>         | 3         |
| <b>Packing Group</b>        | III       |
| <b>IATA</b>                 |           |
| <b>UN-No</b>                | UN1105    |
| <b>Proper Shipping Name</b> | PENTANOLS |
| <b>Hazard Class</b>         | 3         |
| <b>Packing Group</b>        | III       |
| <b>IMDG/IMO</b>             |           |
| <b>UN-No</b>                | UN1105    |
| <b>Proper Shipping Name</b> | PENTANOLS |
| <b>Hazard Class</b>         | 3         |
| <b>Packing Group</b>        | III       |

## 15. Regulatory information

### International Inventories

| Component            | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1-Butanol, 3-methyl- | 123-51-3 | X   | -    | X    | ACTIVE                                        | 204-633-5 | -      | -   |

| Component            | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------|----------|-------|----------|------|------|------|------|-------|-------|
| 1-Butanol, 3-methyl- | 123-51-3 | X     | KE-23575 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component            | CAS-No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| 1-Butanol, 3-methyl- | 123-51-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component            | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities | Seveso III Directive (2012/18/EC) - Qualifying Quantities | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------|----------|-----------------------------------------------------------|-----------------------------------------------------------|----------------------------|------------------------------------|
| 1-Butanol, 3-methyl- | 123-51-3 |                                                           |                                                           |                            |                                    |

|                      |          | for Major Accident Notification | for Safety Report Requirements |                |                |
|----------------------|----------|---------------------------------|--------------------------------|----------------|----------------|
| 1-Butanol, 3-methyl- | 123-51-3 | Not applicable                  | Not applicable                 | Not applicable | Not applicable |

## 16. Other information

|                  |                                                                                                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared By      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                 |
| Creation Date    | 13-September-2010                                                                                                                                                                  |
| Revision Date    | 28-December-2021                                                                                                                                                                   |
| Print Date       | 28-December-2021                                                                                                                                                                   |
| Revision Summary | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**